

Measurement Invariances of the PHQ-Across Gender and Age

Introduction

The Patient Health Questionnaire-9 (PHQ-9) is a validated instrument used to measure depression severity. It has been extensively validated in various adult populations and translated into various languages.

However, there is limited research on its use with adolescent samples. In addition, its factor structure consistency and measurement invariance across genders and age subgroups need to be clarified.

This study aims to assess the usefulness of the PHQ-9 by examining its factor structure using a large community sample of adolescents in Hong Kong. The study seeks to confirm the single-factor structure of the PHQ-9 and examine its factorial invariance across genders and different age subgroups during adolescence.

Methods

The study analyzed secondary data from a cross-sectional survey exploring adolescent mental health risks and protective factors. The participants were students from 16 secondary schools from two districts in Hong Kong, randomly selected and recruited between 2011 and 2014(Allgaier et al., 2012). After written informed consent was obtained from both the students and their parents, the students anonymously completed the self-reported questionnaire. Trained research assistants surveyed a classroom with no teachers. The study was approved by the

Human Ethics Committee of the participating university, and it was conducted according to the Declaration of Helsinki.

The sample for the study comprised 10,933 adolescents with an average age of 14.8 years ($SD = 1.6$). The sample consisted of 4847 males, 6038 females, and 48 respondents who did not provide information about their gender. Students who could not speak and read Chinese were excluded (Burdzovic et al., 2017).

Measures

The study utilized several measures, including the PHQ-9 to measure the presence and severity of states of depression, the GAD-7 to measure anxiety, the RSES to measure self-esteem, and the Mastery Scale to assess perceived control. The psychometric properties of these measures were previously established in both adult and adolescent samples. The article describes a study that measured depression, anxiety, self-esteem, and perceived control in a sample of Chinese adolescents (Galenkamp et al., 2017). Depression was measured using the PHQ-9, which consists of nine items assessing the frequency of depressive symptoms in the past two weeks. Anxiety was measured using the GAD-7, which consists of seven items assessing the frequency of generalized anxiety symptoms. Self-esteem was measured using the RSES, which consists of ten items assessing the level of self-esteem. Perceived control was assessed using the Mastery Scale, which consists of seven items measuring a person's feeling of control over life events. The psychometric properties of these measures were previously validated in Chinese adolescent samples and were found to be reliable and valid in the present study. The study aimed to explore the relationships between these variables and how they may affect each other in Chinese adolescents.

Data analysis

The passage describes the evaluation of the PHQ-9, a questionnaire used to assess depression severity. The evaluation involves various statistical analyses to test the questionnaire's factor structure, reliability, and construct validity.

Confirmatory factor analyses (CFA) were conducted to fit a single-factor model to the PHQ-9 items. The model was first fitted without allowing for any residual correlations among items, and then LM tests were used to identify item pairs that would improve the fit of the model if they were allowed to correlate. Multiple-group CFAs were also performed separately by gender and age to examine measurement invariance(Bentler, 2006). The model's fit and data were evaluated using various fit indices, including SRMR, robust CFI (R-CFI), and robust RMSEA (R-RMSEA).

To assess the internal consistency of the PHQ-9, Cronbach alpha was used, and a value of $>.7$ was considered a criterion of good reliability. Finally, construct validity was evaluated using Pearson correlation coefficients between the PHQ-9 score and relevant measures, including GAD-7, RSES, and Mastery(Belfer, 2008). It was hypothesized that PHQ-9 scores positively correlate with GAD-7 scores and negatively correlate with RSES and Mastery scores.

Conclusion

To address this limitation, this study was conducted in a specific cultural context (i.e., Hong Kong), and the generalizability of the findings to other cultural contexts is still being determined. Finally, this study did not examine the test-retest reliability of the PHQ-9, an important psychometric property to assess the stability of the measure over time.

In summary, the results of this study suggest that the PHQ-9 has a similar one-factor structure for both males and females and for all six age groups of Chinese adolescents, indicating that the measure has measurement invariance across gender and age subgroups. The findings provide important information on the psychometric properties of the PHQ-9 in the context of adolescent depression screening and assessment (Chen et al., 2017). Further studies are needed to examine the criterion validity, predictive power, and test-retest reliability of the measure and its generalizability to other cultural contexts.

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